SYRIAN PRIVATE UNIVERSITY · GRADUATION PROJECT · SEMINAR I

BIAVORA

A Governed Agentic Business Intelligence Platform

From Heterogeneous Business Data to Trusted Insights, Controlled AI Actions, and Informed Decision-Making

01	02	03
Business data	**Trusted analysis**	**Decision Support**

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BI Voice Agent

Voice Input

Captures spoken language queries from the user.

Speech-to-Text

Transcribes audio input into written text format.

Intent Understanding

Analyzes text to discern the user's specific request or goal.

SQL Generation

Constructs precise database queries based on identified intent.

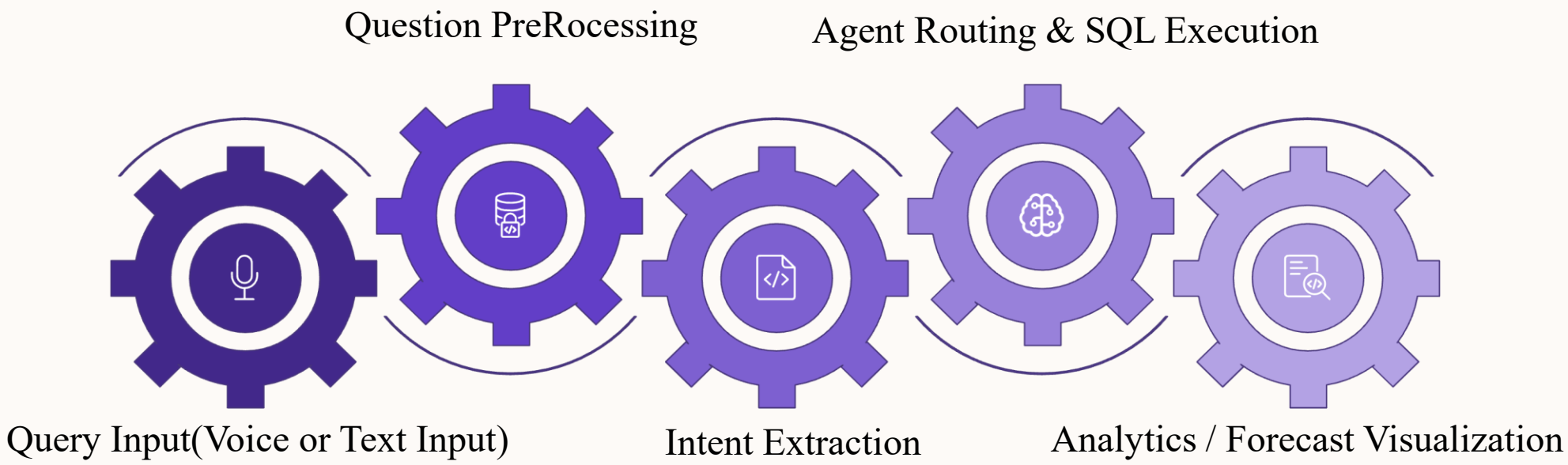
Chart Visualization

Transforms query results into dynamic, readable charts.

Dashboard Output

Displays interactive charts and insights on a user-friendly dashboard.

BI Agentic Platform Overview?



The system processes user queries through intelligent AI and data layers to deliver advanced analytics and enterprise services.

PROBLEM

The Problem: Business Data Exists, but Trusted Insight Is Still Difficult

01 Fragmented Data :

Heterogeneous sources and continuously changing operational data.

02 Inconsistent Semantics :

The same KPI can be defined differently across reports and teams.

03 Unreliable AI Analysis :

AI-generated SQL and insights may be plausible but incorrect or unsupported.

04 Fragmented Decision Workflow :

Analysis, actions, governance and executive reporting are disconnected.

⚠ Businesses need continuously synchronized data, governed business definitions, explainable AI analysis, and decision-ready outputs.

State of the Art: AI Is Rapidly Entering Business Intelligence

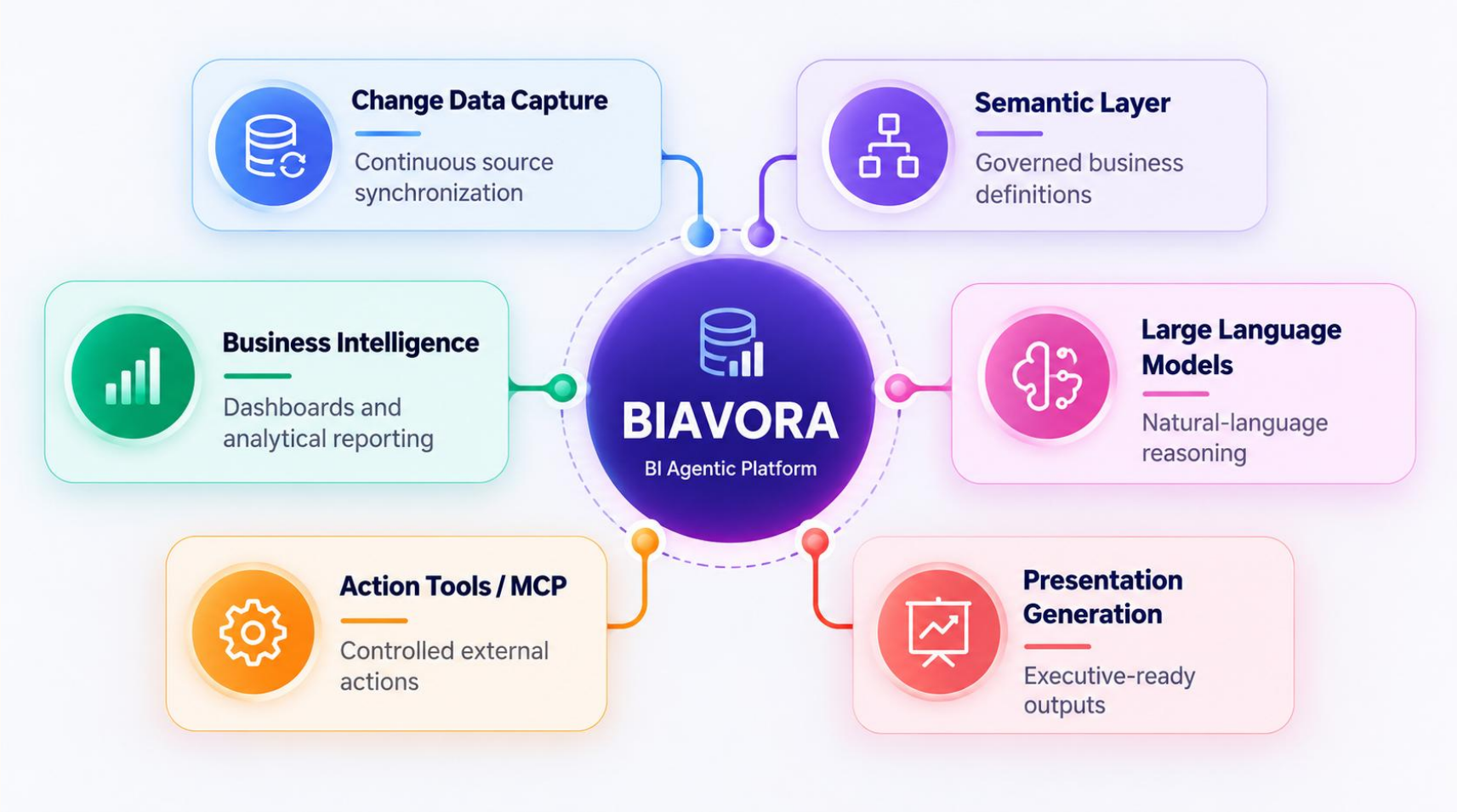
Platform	AI capability and context
Power BI Copilot	Natural-language analytics and AI-assisted reporting grounded in prepared semantic models within the Microsoft Fabric ecosystem.
Tableau Agent / Pulse	Conversational exploration across governed metrics, trends, contributors, correlations, and outliers.
ThoughtSpot Spotter	Agentic analytics for data exploration, modeling, and visualization within governed analytical context.
Snowflake Cortex Agents	Semantic-view-based analytics, multi-step orchestration, and governed tools; Cortex Analyst serves as the structured-data analysis tool.
dbt Semantic Layer	Centralized, reusable, governed business metrics powered by MetricFlow.

BIRD and Spider 2.0 demonstrate that real-world Text-to-SQL remains challenging because enterprise workflows involve large schemas, metadata, multiple SQL dialects, and multi-step reasoning.

INTEGRATION GAP

Research & Engineering Gap

The individual technologies already exist. The challenge is integrating them safely and coherently.



PROPOSED ARCHITECTURE

BIAVORA: Proposed Solution



Query Service is the sole analytical execution authority.

Multi-tenant isolation · Provider-neutral AI · Governed metrics · Evidence · Human approval · Production-oriented security

CURRENT IMPLEMENTATION

Current Progress: The Core Foundation Is Already Implemented

COMPLETED

Query Security Boundary

Tenant-aware authorization, resource limits, fail-closed execution and controlled analytical authority.

COMPLETED

External Boundaries

Secrets, secure uploads, subscription/payment containment and security regressions.

LOCAL ENGINEERING CLOSED

Data Integration Foundation

Persistent sources/datasets, encrypted credentials, secure transport, rollback, tenant isolation and green release CI.

CURRENT · Foundation-D — PostgreSQL CDC. External production deployment validation remains deferred to Production Readiness; it is not complete.

PLANNED · ALL CAPABILITIES BELOW

Planned Innovations

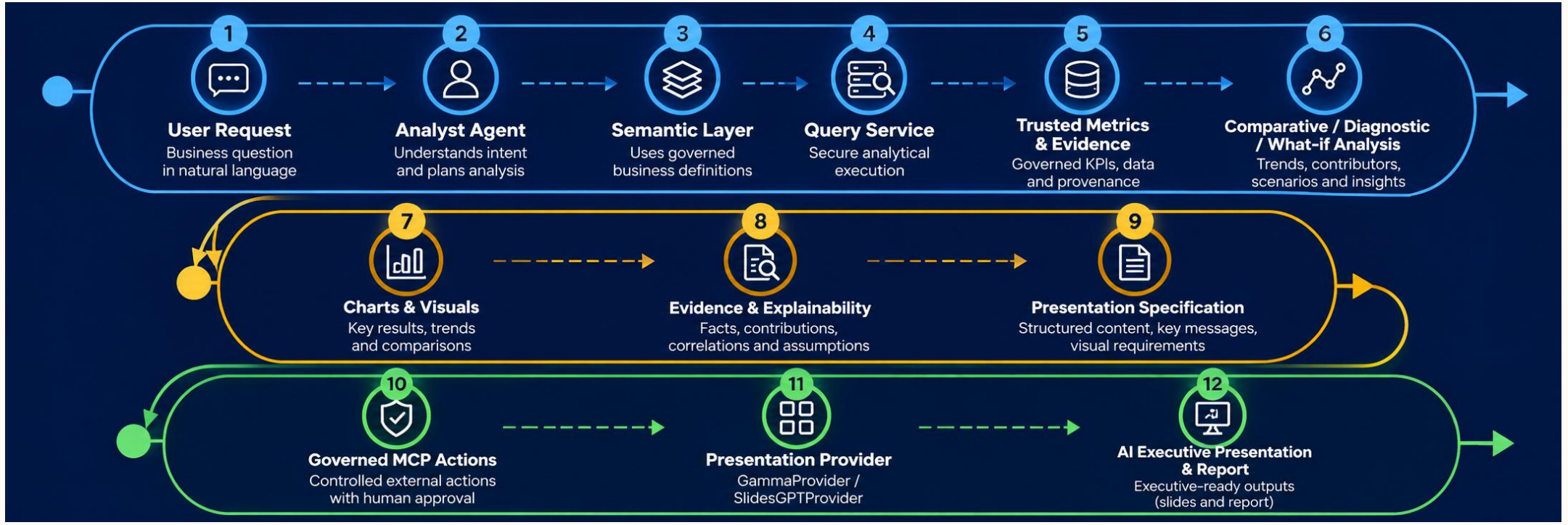
Capability layer	Planned scope
Executive delivery	AI Presentation & Executive Report Generation through a provider-neutral PresentationProvider, with GammaProvider and SlidesGPTProvider.
Trust & controlled actions	Trust & Explainability, evidence/provenance, governed MCP actions, human approval, AI/security evaluation, and cost-aware metering.
Agentic analytics	Governed Analyst Agent, multi-step analysis, What-if Analysis, predictive support, and a limited Data Reliability Agent.
Semantic intelligence	Governed Semantic Layer & Metrics, reusable business definitions, and a human-reviewed Semantic Modeling Agent.
Real-time data foundation	Existing multi-database integration, PostgreSQL CDC, Debezium, Kafka Connect, Kafka-based event flow, and continuous synchronization to analytical storage.

Capability map reads from the data foundation upward to executive delivery. These are planned capabilities, not completed functionality.

PLANNED SCENARIO

Example: From a Business Question to an Executive Presentation

> “Build a presentation explaining this year's losses, the major contributors, trends, risks, and recommended actions.”



BIAVORA performs the governed analysis; the external provider only composes the final presentation. Analytical outputs distinguish observed facts, contributions, correlations, likely drivers, and causal claims.

DELIVERY SEQUENCE · 2026

Implementation Roadmap — Part I

COMPLETED EARLY

1 Sprint 0 : 27–28 Oct 2026

Foundation Closure

Foundation-C local closure, release CI GREEN, security regression validation, and PostgreSQL CDC readiness.

IN PROGRESS

2 Sprint 1 : 28 Oct–16 Nov

PostgreSQL CDC

PostgreSQL WAL, Debezium, Kafka Connect, consistent snapshot-to-stream transition, INSERT/UPDATE/DELETE synchronization, replay, recovery, and reconciliation.

PLANNED

3 Sprint 2 : 17–29 Nov

Semantic Layer & Metrics

Business entities, dimensions, measures, relationships, governed metrics, semantic validation, and versioned publication.

PLANNED

4 Sprint 3 : 30 Nov–6 Dec

Semantic Modeling & Governance

Semantic Modeling Agent, model proposals and validation, human-reviewed publishing, evidence foundation, MCP policy, and metering foundations.

PLANNED · ALL SPRINTS BELOW

Implementation Roadmap — Part II

Sprint 4 · 7–15 Dec — Governed Analyst Agent, What-if Analysis, limited Data Reliability, Trust & Explainability, evidence, and governed MCP approvals..

Sprint 5 · 16–23 Dec — AI Presentation & Executive Report Generation, PresentationProvider with GammaProvider and SlidesGPTProvider, narrative, charts, and export.

Sprint 6 · 24–29 Dec — System integration, failure/restart recovery, security convergence, metering reconciliation, and end-to-end acceptance.

Sprint 7 · 30 Dec–4 Jan 2027 — AI evaluation, factuality, security testing, performance hardening, and release-quality validation.

Sprint 8 · 5–9 Jan — Release candidate: full regression, staging validation, backup/restore, migration/rollback, and production-readiness verification..

Target public release: 10 January 2027

Planned target, not a claim of completed production readiness.

BIAVORA's Contribution



Common Approach	BIAVORA Target
Raw NL → generated SQL	Semantic model + governed Query
Static data snapshots	Snapshot + continuous CDC
Different KPI definitions	Governed reusable metrics
Single AI response	Multi-step Analyst Agent
Opaque AI answer	Evidence-backed explanation
AI directly executes tools	Policy + Human Approval
Manual executive reporting	Automated evidence-based presentation
Vendor-specific implementation	Provider abstraction
Simple request quotas	Resource/cost-aware metering

Evaluation and Expected Outcome

Data & CDC

- Snapshot/stream consistency
- INSERT/UPDATE/DELETE correctness
- Restart/replay/recovery
- Source-to-destination reconciliation

Analytical AI

- Intent classification
- Semantic metric selection
- SQL execution correctness
- Tool selection
- Hallucination/factuality evaluation

Trust & Security

- Tenant isolation
- Unauthorized data-access tests
- Prompt/tool injection tests
- Evidence completeness
- Human approval enforcement\

System Quality

- Latency
- Throughput
- Reliability
- Resource usage
- Provider cost/usage

THANK YOU FOR LISTENING!

